When Memory Enters an ODE

Solving a Highly Oscillatory Volterra Integro-Differential Equation with SIRPY

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SIRPY · Problem 13 — The Integral That Wasn't There

An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.

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A hard integral, sixteen times over

$$y' + y = \int_0^x \sin(100(x-t))\, y(t)\, dt, \qquad y(0) = 1$$

This is an integro-differential equation with a **convolution memory kernel**. The derivative of y at any point depends on an integral of y over its entire history — and that history is weighted by $\sin(100(x-t))$, which oscillates with a period of about 0.063[1].

Across the interval [0, 1], that kernel completes roughly **sixteen full oscillations**[2]

That is what makes the problem unpleasant. The conventional approach is to evaluate the convolution numerically at every step of the march, and oscillatory quadrature is precisely where numerical integration performs worst. A rapidly alternating integrand produces large positive and negative contributions that must cancel to leave a small result, so you need many sample points per period simply to avoid the cancellation error swamping the answer. Do that

[1]The general form is $sin(\omega\, z)$ where $omega = 100$ and $z = x - t$.

$$\text{Period} = 2\pi/\omega \;=\; 2\pi/100 \approx 0.063$$

[2]oscillations = 1/period = 1/0.063 ≈ 16.

at every step of an ODE march, over sixteen periods, and the cost and the error accumulate together.

Kernels of this form appear wherever a system's present rate depends on an oscillating memory of its past: viscoelastic materials under cyclic loading, feedback control with delayed and resonant response, wave scattering with periodic structure.

What we handed the solver

```
solve_integro(order=1,
              phi_str="-y",
              kernel_str="sin(100*(x-t))*y",
              ic={0: 1}, x0=0, x1=1.0)
```

The equation, as written. The forcing term, the kernel, the initial condition, the interval. No quadrature rule chosen, no sample density specified, no mesh, no tuning.

What came back

Start with the last line of the report, because it is the whole story:

```
INFO: the convolution kernel sin(100*(x-t))*y satisfies a linear
      relation of order 2, so the integral was removed exactly by
      carrying 2 auxiliary variable(s). The problem became a
      first-order system of 3 equations with no integral term.
```

SIRPY did not compute the integral. It **eliminated** it.

The observation is that $\sin(100(x-t))$ satisfies a second-order linear differential relation. Because of that, the convolution can be differentiated into a closed system: introduce the sine convolution and its cosine partner as two new unknowns, and the integral term disappears entirely, replaced by two extra differential equations.

The original problem — one equation with an oscillatory memory integral — becomes three ordinary differential equations with **no integral anywhere**. And the reduction is exact. Nothing is approximated, sampled, or discretized in the process.

The hardest part of the problem was not solved efficiently. It was removed.

Solved and verified

```
Piecewise power-series solution (46 segments of degree 20 covering [0, 1])

INFO: the interval [0, 1.0] exceeds a single series' reliable range;
      the solution was continued analytically over 46 segments
      (piecewise power series - still fully analytic on every piece).

INFO: verification by substitution - VERIFIED: every equation of the
      system is satisfied to 8.65e-10 on [0, 1] (exact coefficient
      differentiation, interior points). Also checked: initial data
      to 0.00e+00.
```

Forty-six analytic segments across [0, 1] — roughly three per oscillation of the kernel — with every equation of the resulting system verified by substitution to 8.65×10^{1}, and the initial condition met exactly.

The result is a function, not a table: readable, differentiable, and evaluable anywhere on the interval.

Kernel oscillations across [0, 1]	$\approx$ 16
Quadratures performed	**0**
Auxiliary variables carried	2
Resulting system	3 first-order equations, no integral
Analytic segments	46
Verification residual	8.65×10^{-10}
Initial data	exact

Independently checked

Because the reduction is exact and the resulting system is linear with constant coefficients, this problem has a closed-form answer reachable by matrix exponential — entirely outside SIRPY. We computed it that way as an independent check:

x	reference y(x)
0.10	0.905800431099483
0.25	0.780762831368421
0.50	0.609596076551990
0.75	0.475960020101381
1.00	0.371625737511750

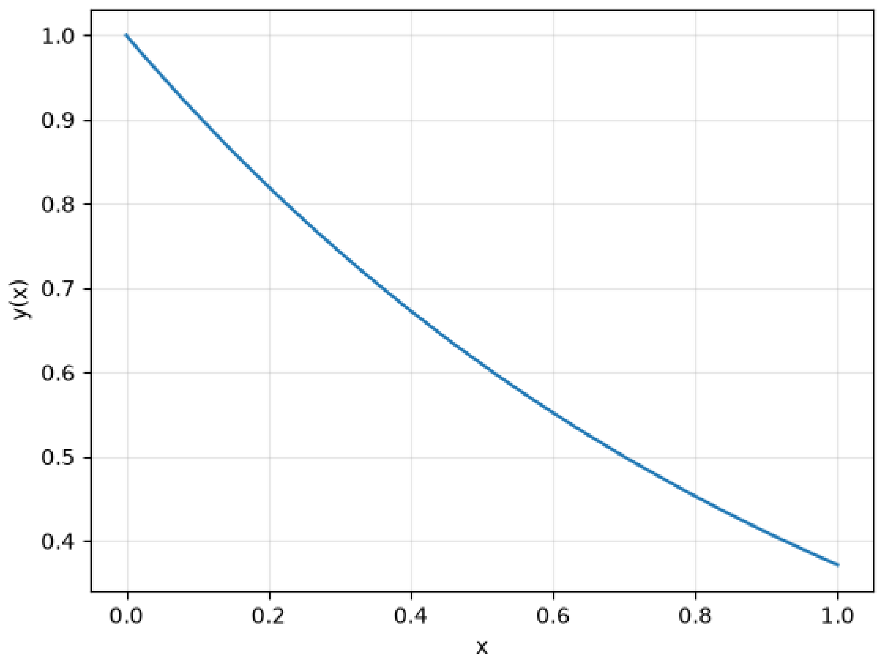

Figure 1: Solution

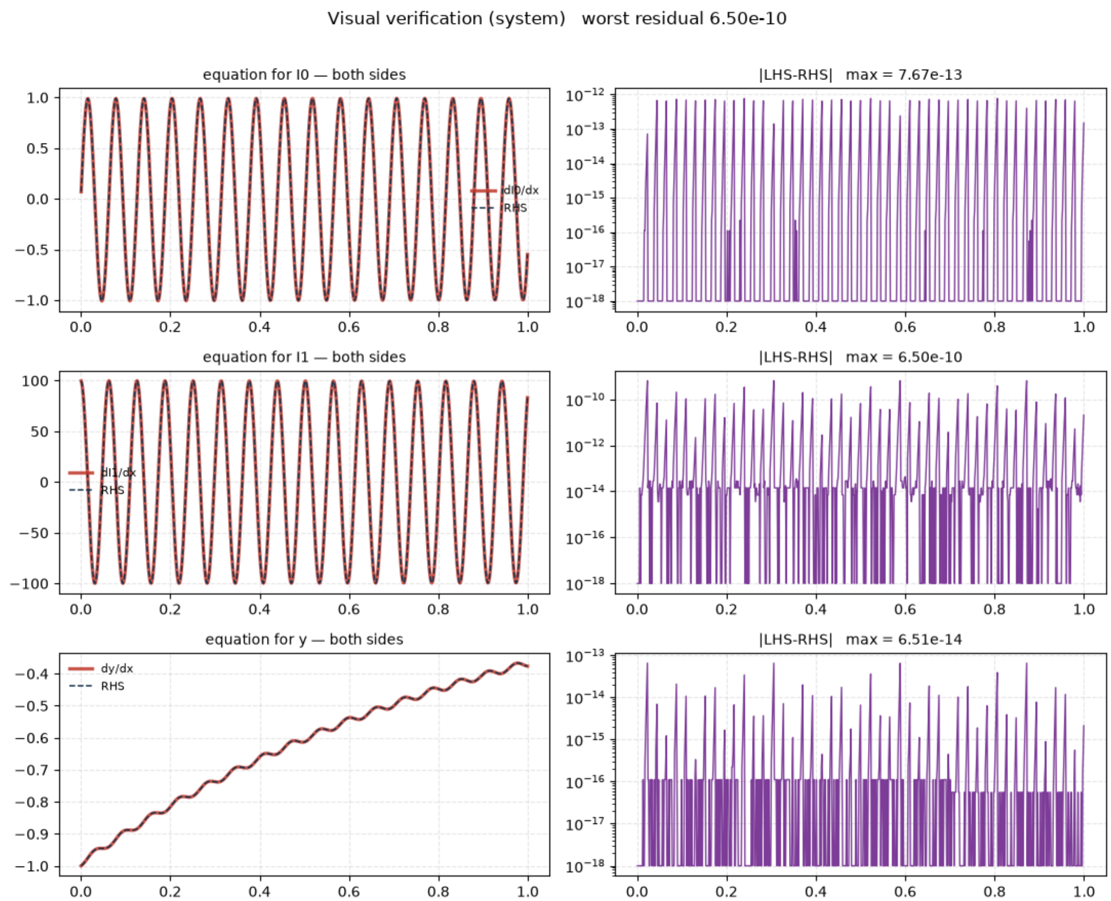

Figure 2: Verification: both sides of the equation compared across the interval. $I0$ and $I1$ are the two auxiliary variables

What this problem shows

- **The equation was entered as written.** Kernel, forcing term, initial condition, interval. No quadrature scheme, no sample density, no mesh.
- **The integral was eliminated, not approximated.** SIRPY recognized that the kernel satisfies a linear relation, carried two auxiliary variables, and reduced the problem to a system with no integral term at all — exactly, with no quadrature performed anywhere.
- **The oscillation stopped being an obstacle.** Sixteen periods of a rapidly alternating kernel would ordinarily demand dense sampling and invite cancellation error. Removed algebraically, they demand neither.
- **It said what it did.** The transformation is reported in the output — what was recognized, how many auxiliary variables were carried, and what the problem became.
- **Verified by substitution**, every equation of the resulting system, to 8.65×10^{-10}, with initial data exact.

The problem library

This entry is part of the SIRPY Problem Library — an ongoing, public catalog of mathematical systems solved and verified by SIRPY.

- **GitHub** — github.com/Mitra-Institute-of-Education-MIE/sirpy-problem-library
- **Web** — mitrainstituteofeducation.org/sirpy

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SIRPY is in final validation prior to its first public release. Every line quoted above is verbatim from the solver's output; nothing is reproduced from memory.